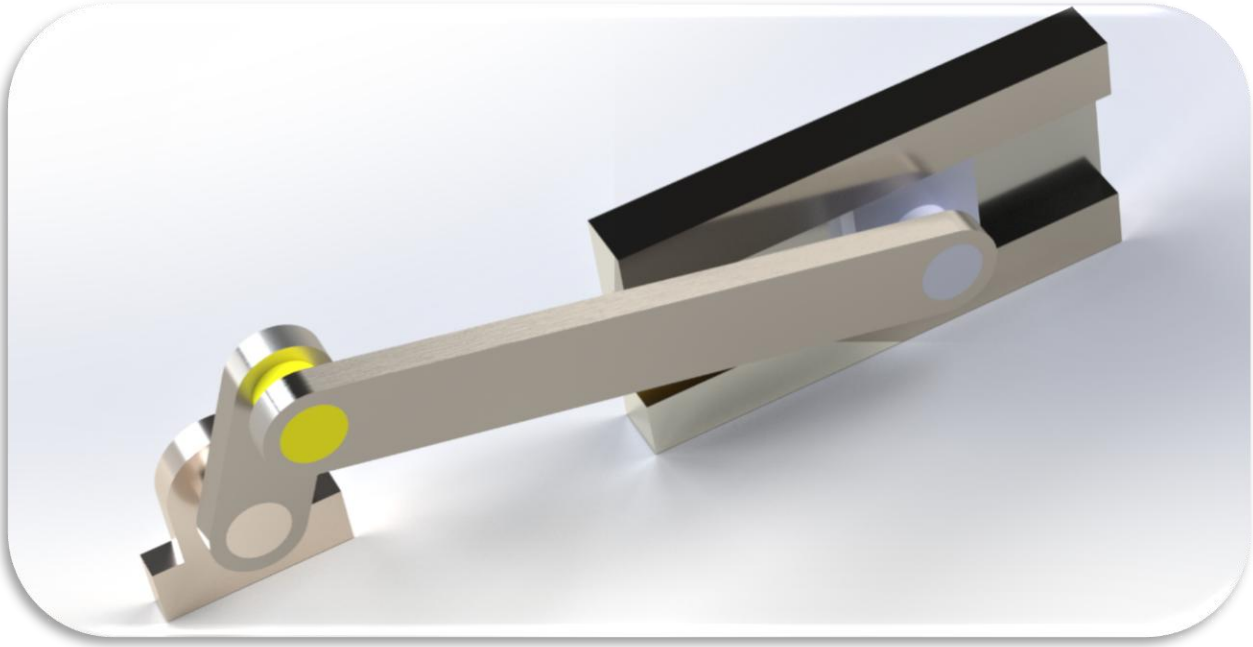
Program : Mechanical Design & Simulation Internship — CodeAlpha

-Task 3 : Motion Simulation
-Software : SolidWorks 2025



Period : June 10 – July 10

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CODE ALPHA

1. Task Objective

This study is part of the CodeAlpha mechanical design internship. The objective is to model a motion-conversion mechanism, to carry out its complete dynamic simulation in SolidWorks Motion Analysis, and to extract the kinematic data and joint reaction loads for the subsequent strength-of-materials sizing phases.

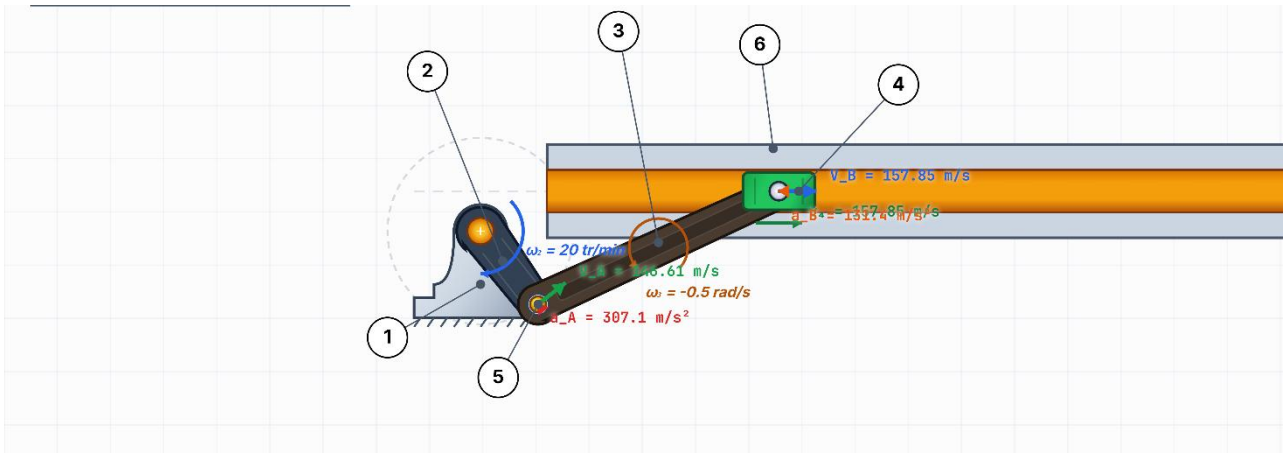
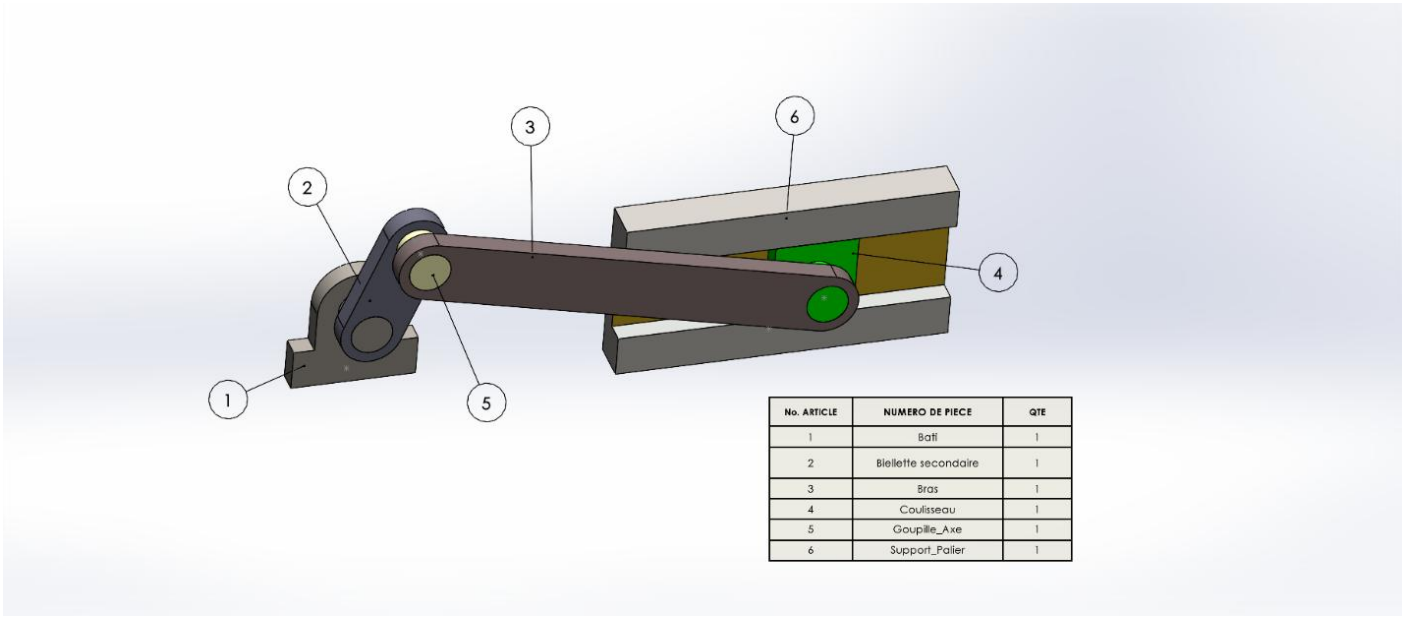


Figure 1: The joystick-cursor system to be modeled

2. Description of the Mechanism

The selected system is a slider-crank mechanism, a classic architecture that converts continuous rotation into reciprocating translation. The 3D model comprises four distinct kinematic components:



No. ARTICLE	NUMERO DE PIECE	QTE
1	Bati	1
2	Biellette secondaire	1
3	Bras	1
4	Coulisseau	1
5	Goupille_Axe	1
6	Support_Palier	1

Figure 2: 3d modeling of the system | SolidWorks

Dimensional note: The crank ratio (crank radius / connecting rod length) was sized to guarantee full rotation without jamming or kinematic lock-up.

3. Methodology and Simulation Hypotheses

3.1 Simulation Environment

The simulation is carried out in SolidWorks Motion Analysis (Adams solver), providing a rigorous rigid-body dynamics computation that accounts for the inertia matrices automatically generated through material assignment.

3.2 Adopted Hypotheses

- ✓ Material assignment — actual mass properties applied to each moving part
- ✓ Ideal joints : pin joints at the articulations, prismatic joint for the slider guidance
- ✓ No-load operation : no external resisting force or payload on the slider
- ✓ Ideal drive : rotary actuator at constant angular velocity (20 rpm)
- ✓ Rigid bodies : elastic deformations of the parts neglected
- ✓ Frictionless : joints assumed to be dissipation-free
- ✓ Absolute reference frame : frame fully fixed, zero degrees of freedom

No. ARTICLE	NUMERO DE PIECE	Material	QTE
1	Bati	S235JR	1
2	Biellette secondaire	C45	1
3	Bras	AL 6061	1
4	Coulisseau	C45	1
5	Goupille_Axe	Acier allie	1
6	Support_Palier	S235JR	1

Figure 4: Choice of materials

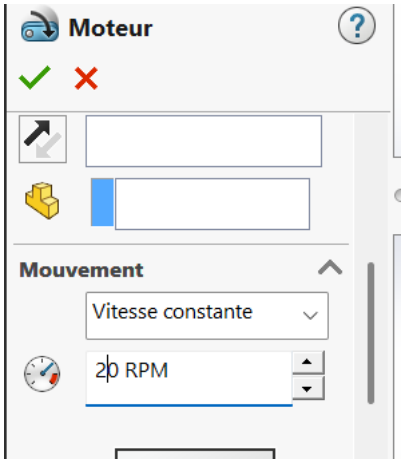


Figure 3: Motor configuration

4. Results and Interpretations

The solver confirms the overall consistency of the model: the theoretical period of 3 s (N = 20 rpm → T = 60/20) is scrupulously respected across all readings, with each pattern repeating at t = 0 s, 3 s, and 6 s. The three analyses below summarize the results obtained.

Verified cycle period	Max. slider accel.	Max. rod angular velocity	Max. reaction force
T = 3 s N = 20 rpm → T = 60/20	**|0.30|** m/s² (negative peak at BDC)	**~30** deg/s (horizontal axis)	**0.021** N (crank–rod joint)

4.1 Linear Kinematics of the Slider

The linear kinematics of the slider reveals the **π/2** phase shift characteristic of slider-crank systems: the acceleration reaches its extrema exactly at the instants where the velocity vanishes (top and bottom dead centers). A notable asymmetry is observed between the negative acceleration peak **(≈ −0.30 m/s²)** and the positive peak **(≈ +0.20 m/s²).** This distortion of the kinematic sinusoid — not an anomaly but a physical signature — is directly proportional to the crank ratio **λ = r/L** and reflects the obliquity of the connecting rod during rotation. Its presence validates the accuracy of the solver against the non-linear equations of motion.

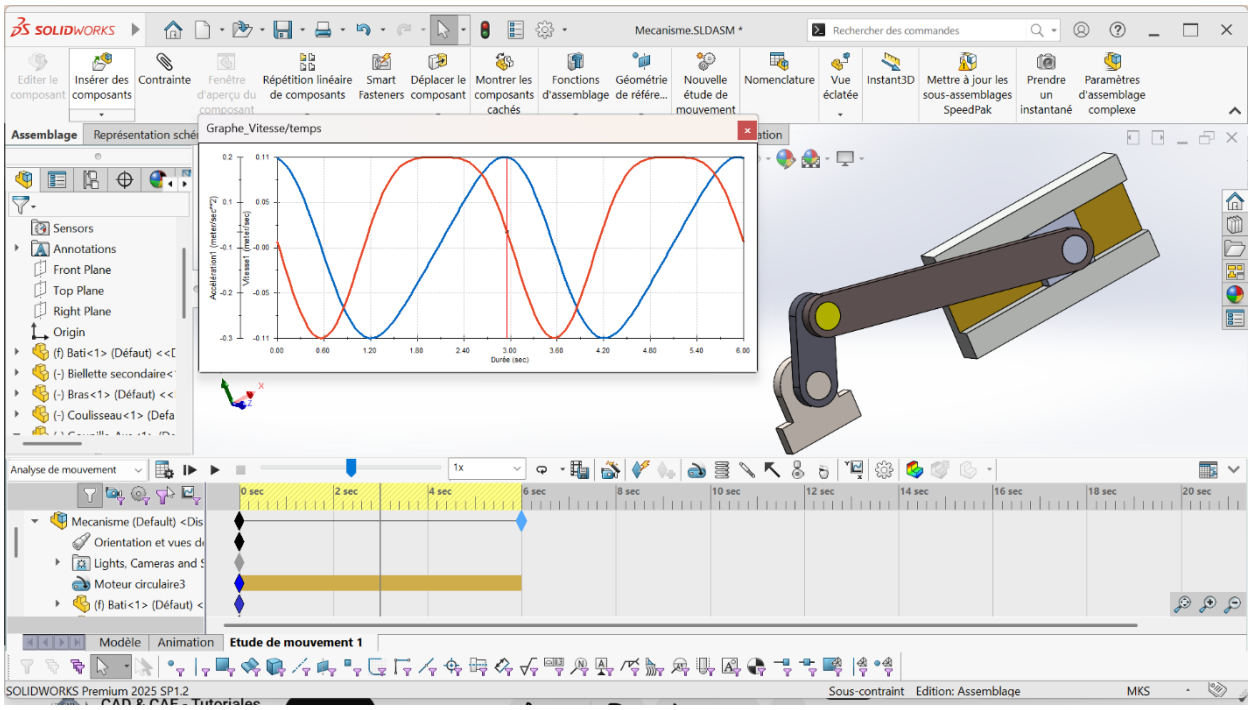


Figure 5: Linear kinematics

Linear velocity

Linear acceleration

4.2 Angular Kinematics of the Connecting Rod

The angular velocity and acceleration curves of the connecting rod are displayed as magnitudes (vector norms), which justifies their remaining strictly in the positive domain. The angular velocity vanishes when the rod reaches its maximum inclination (reversal of the oscillation direction); it is precisely at these instants that the angular acceleration peaks at ≈ 85 deg/s². Conversely, as the crank passes through the horizontal axis, the angular velocity is at its maximum (≈ 30 deg/s) while the acceleration passes through zero, in accordance with the reference analytical kinematic derivation.

Angular velocity

Angular acceleration

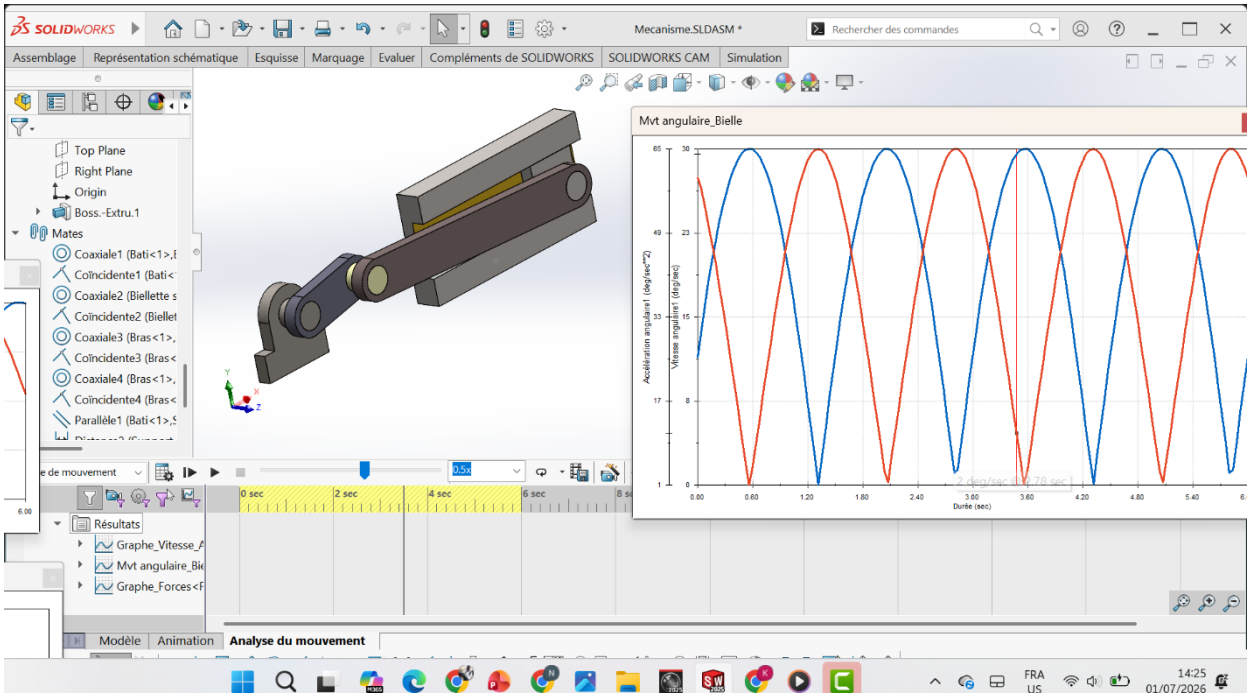


Figure 6: Angular Kinematics | SolidWorks

4.3 Internal Dynamics - Reaction Loads at the Pin Joints

Under no-load conditions, the reaction forces correspond to pure inertia forces, consistent with the strict application of the Fundamental Principle of Dynamics (Newton's 2nd law). The table below summarizes the loads measured at the two critical joints:

Joint	Max. reaction force	Physical interpretation	Synchronization
Crank–rod pin joint (main axis)	≈ 0.021 N	Carries the cumulative inertia: rod mass + slider mass	Synchronous with acceleration peaks
Rod–slider pin joint (secondary axis)	≈ 0.015 N	Carries only the inertia of the slider itself (downstream)	Synchronous with acceleration peaks

Key observation: The ratio F_main / F_secondary ≈ 1.4 is mechanically consistent with the mass distribution. The rigorous synchronization of the load peaks with the linear acceleration peaks confirms the correct application of the Fundamental Principle of Dynamics by the numerical solver.

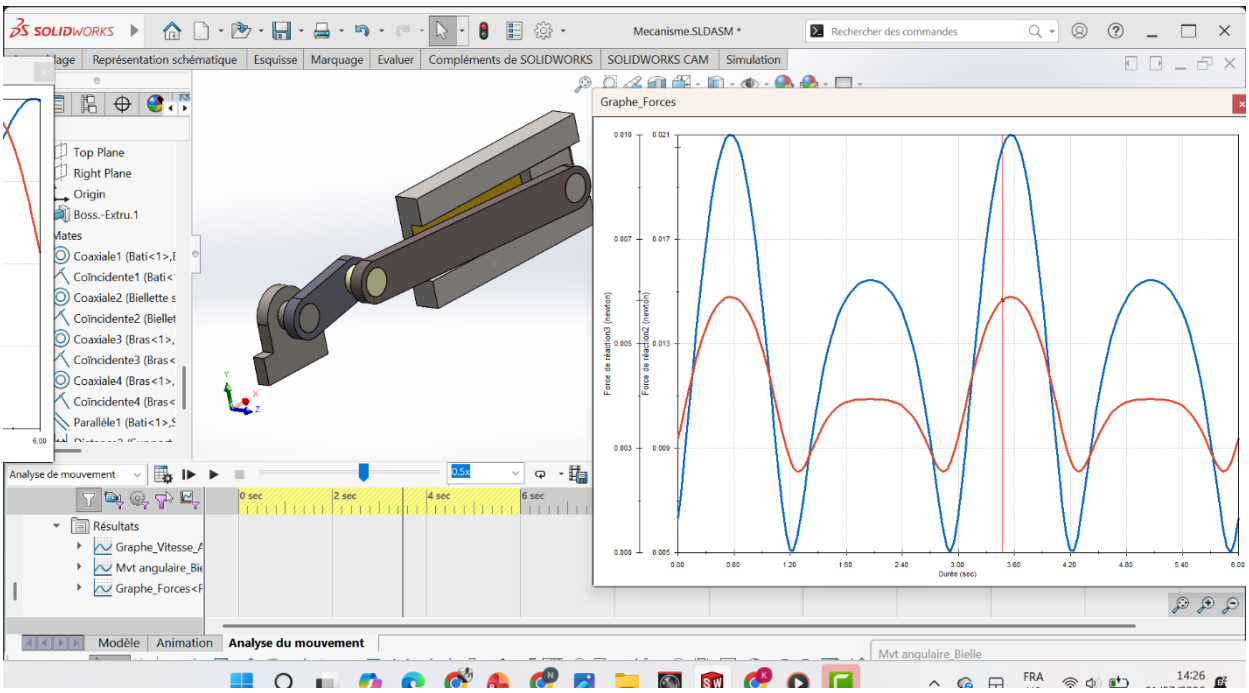


Figure 7: force simulation

Red curve: Joint reaction force : crank–rod (arm) pin joint

Blue curve: Joint reaction force : rod (arm)–slider pin joint

Comparative study : simulation with and without applied materials

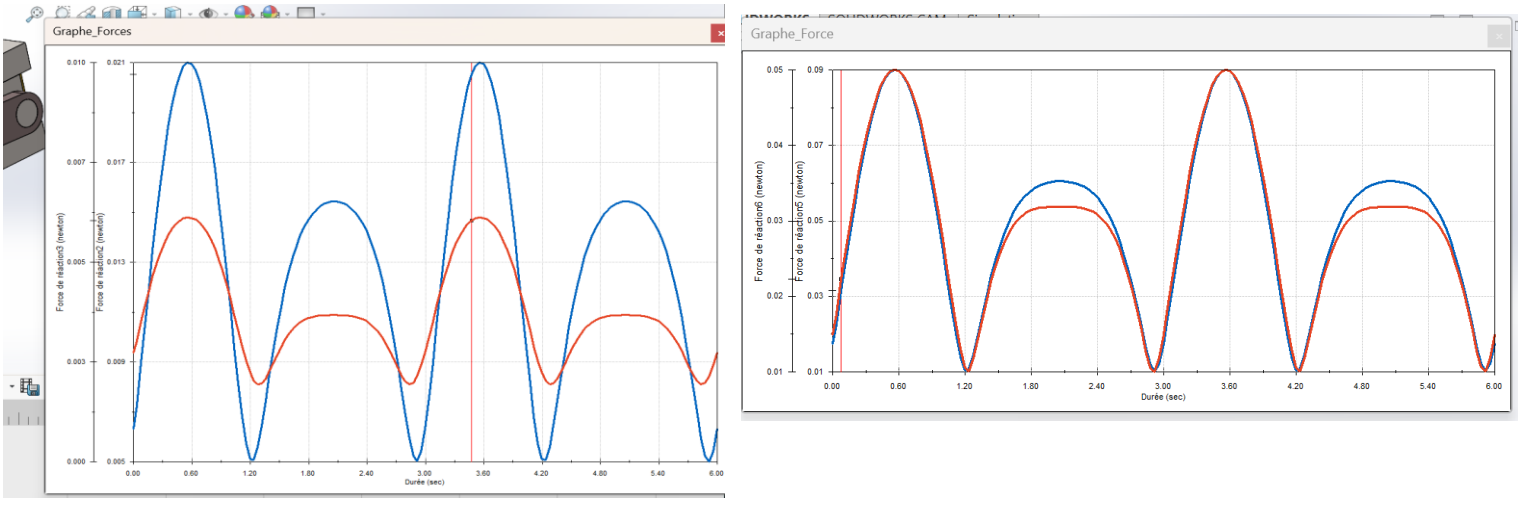


Figure 8: Comparative study : simulation with and without applied materials

The comparative analysis of the results highlights the influence of component mass on the dynamic behavior of the mechanism. Although the kinematics remains unchanged, as it is imposed by the motor, the increase in mass leads to a significant increase in the loads transmitted to the joints. The maximum force exerted at the pin joint thus rises from **0.021 N to 0.09 N** after the assignment of a denser material. This evolution is consistent with the fundamental principle of dynamics, expressed by the relation $\sum \boldsymbol{F} = \boldsymbol{ma}$, which indicates that, at constant acceleration,

inertia forces are directly proportional to mass. These results confirm the importance of optimizing the mass of the moving elements in order to reduce mechanical stresses, improve the dynamic performance of the system, and extend the service life of the components.

5. Summary and Contributions of the Study

This dynamic simulation constitutes a pivotal step in the product design process. The main technical and methodological contributions are summarized below:

- ✓ Kinematic validation : viability of the slider-crank architecture confirmed without physical prototyping
- ✓ Mastery of the acceleration asymmetry : analytically correlated to the crank ratio **λ = r/L**
- ✓ Mapping of inertia loads : joint reaction loads quantified, providing a basis for strength-of-materials sizing
- ✓ Motion Analysis proficiency : full setup: mates, actuators, post-processing
- ✓ Rigorous interpretation : critical analysis of numerical outputs (magnitudes vs. components)

6. Summary of Results

Parameter	Simulated value	Theoretical expectation	Validation
Cycle period	T = 3.00 s	T = 60/20 = 3 s	✓ **COMPLIANT**
Linear accel. peak (negative)	−0.30 m/s²	Expected asymmetry (λ)	✓ **COMPLIANT**
Linear accel. peak (positive)	+0.20 m/s²	|peak+| < |peak−| (λ)	✓ **COMPLIANT**
Max. rod angular velocity	≈ 30 deg/s	Max at horizontal axis crossing	✓ **COMPLIANT**
Max. rod angular accel.	≈ 85 deg/s²	Max at direction reversals	✓ **COMPLIANT**
Main pin joint reaction force	≈ 0.021 N	F = m_rod+slider × a	✓ **CONSISTENT**
Secondary pin joint reaction force	≈ 0.015 N	F = m_slider × a	✓ **CONSISTENT**

GENERAL CONCLUSION

All the results obtained are in perfect agreement with the analytical predictions of mechanism mechanics. The simulation validates the kinematic architecture of the designed slider-crank mechanism and demonstrates proficiency in the SolidWorks Motion Analysis environment. The extracted load data, although obtained under no-load conditions, provide a rigorous foundation for tackling the structural sizing phases under real industrial loads.